CAPSTONE PROJECT - TECHNICAL PRESENTATION

Fraud Detection Using Machine Learning

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Post-Graduate Diploma in AI and Machine Learning
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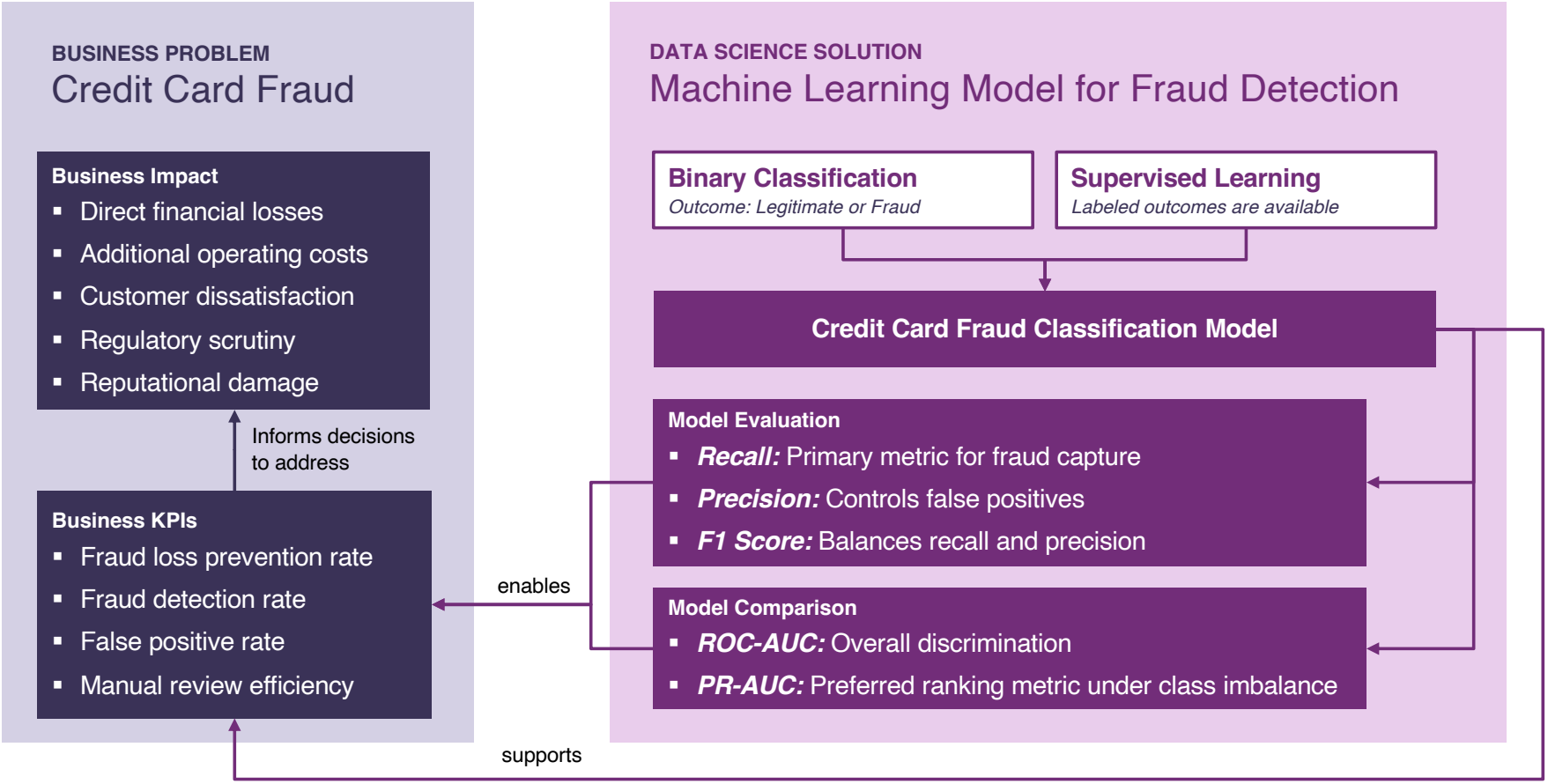
PROBLEM DEFINITION



DATASET OVERVIEW AND EDA

Dataset Description

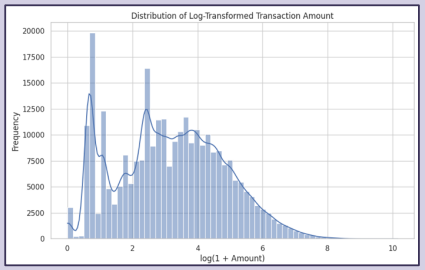
- ***Name:*** Credit Card Fraud Detection
- ***Source:*** Kaggle
- ***Description:*** Real-world European credit card transactions made in two days in September 2013

Variable		Type	Description
Target	**Class**	Integer	Fraud label: 0 = legitimate, 1 = fraud
Predictors	**Time**	Float	Elapsed time in seconds since the first recorded transaction
	Amount	Float	Transaction amount (Euros)
	V1-V30	Float	PCA-transformed features, anonymized for data privacy

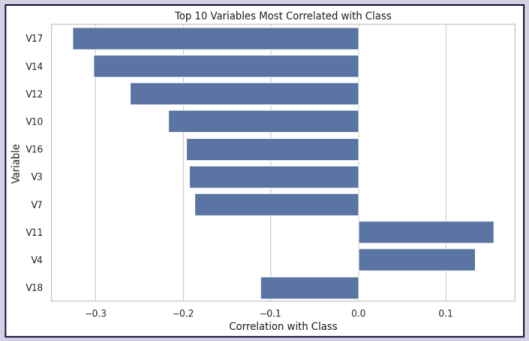
	Check	Result
0	Total rows	284807
1	Total columns	31
2	Total missing values	0
3	Columns with missing values	0
4	Duplicate rows	1081

Database Size
~285K transactions

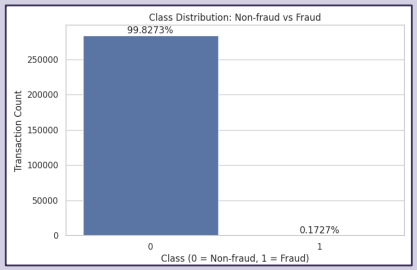
Data Quality
No missing values, ~1K duplicate rows



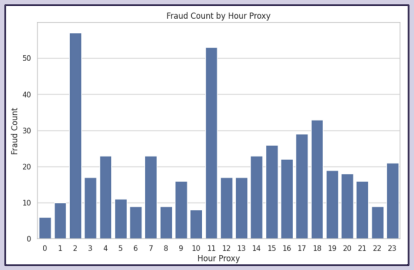
Amount Distribution
Highly right-skewed, with high-value outliers



Correlation Patterns
Some transformed variables (V) are meaningfully correlated with fraud



Class Distribution
Severe class imbalance
(~0.17% fraud)



Time Distribution
Potentially useful time patterns

Implication
Additional feature engineering is required

FEATURE ENGINEERING

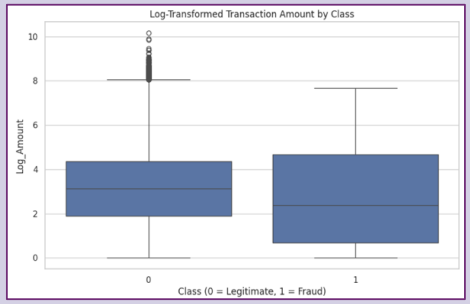
Data Handling

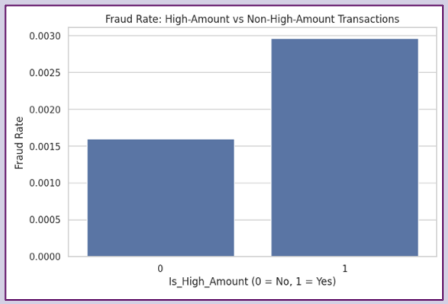
Nulls: None (no missing values)

Outliers: Retained but will be transformed and scaled

Duplicates: Removed from working dataset

Engineered Amount Features





Log_Amount

- Logarithmic transformation of Amount
- Reduces influence of extreme values without removing them

Observations

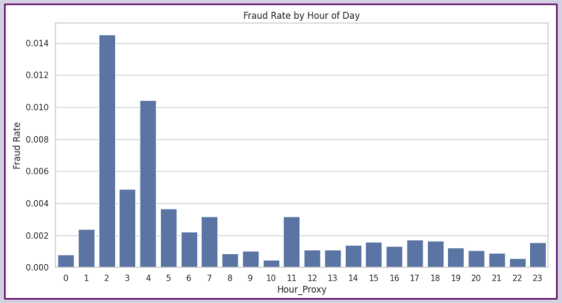
- Fraud transactions more dispersed
- Fraud not just concentrated on large amounts

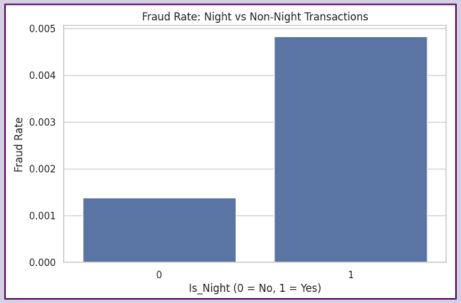
Is_High_Amount

- Binary: 1 if Amount ≥ 95th percentile
- Explores fraud signal of outliers

Observation: Slightly higher fraud rate for high-amount transactions

Engineered Time Features





Hour_Proxy

- Transforms Time into approx. hourly-cycle proxy
- Improves interpretability of timing patterns

Hour_Sin & Hour_Cos

- Transforms Hour_Proxy into a cyclical variable
- Captures 24-hr nature of time

Observation: Fraud rates higher from Hours 1 to 7 and Hour 11

Is_Night

- Binary: 1 if transaction is between midnight (Hour 0) and 5 a.m. (Hour 5)
- Explores fraud signal of unusual timing

Observation: Significantly higher fraud rate for night transactions

DATA PREPROCESSING

Pipeline: Engineered Dataset → Split → Scale → Select → Transform

Train-Test Split

- 80:20 ratio with stratified sampling
- Preserved class distribution

```
X_train shape: (226980, 36)
X_test shape: (56746, 36)
y_train shape: (226980,)
y_test shape: (56746,)

Training set class distribution:
```

Class	Count	Proportion
0	226602	0.9983
1	378	0.0017

```
Test set class distribution:
```

Class	Count	Proportion
0	56651	0.9983
1	95	0.0017

Feature Scaling

- Standard scaling of predictors
- Fit on training set only to prevent leakage

Feature Selection

- Reduced number of predictors from 36 to 20
- Captured both independent and interaction-driven fraud signals

1. Mutual Information

- Filter-based feature selection
- Captures linear & non-linear dependencies

	Feature	Mutual_Information_Score
35	Is_Night	0.0161
34	Is_High_Amount	0.0089
18	V17	0.0080
15	V14	0.0079
11	V10	0.0073
13	V12	0.0073
12	V11	0.0066
17	V16	0.0059
4	V3	0.0047
5	V4	0.0046
10	V9	0.0041
19	V18	0.0040
8	V7	0.0038
32	Hour_Sin	0.0038
33	Hour_Cos	0.0038
3	V2	0.0030
31	Hour_Proxy	0.0025
28	V27	0.0022
22	V21	0.0022
7	V6	0.0021

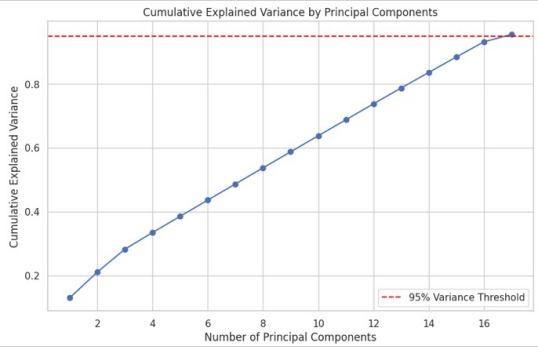
2. Random Forest

- Model-based feature importance
- Captures interactions between variables

	Feature	Importance
11	V17	0.2050
8	V12	0.1291
9	V14	0.1219
6	V10	0.0982
10	V16	0.0807
7	V11	0.0671
5	V9	0.0437
12	V18	0.0380
4	V7	0.0344
2	V4	0.0343
13	V21	0.0278
14	V27	0.0237
1	V3	0.0231
3	V6	0.0212
0	V2	0.0182
16	Hour_Sin	0.0146
15	Hour_Proxy	0.0102
17	Hour_Cos	0.0063
19	Is_Night	0.0012
18	Is_High_Amount	0.0012

Dimensionality Reduction

- Performed Principal Component Analysis (PCA)
- Retained at least 95% of explained variance
- Result: 17 principal components



Datasets for Modeling

- Original feature set (36 features
- Selected feature set (top 20 features)
- PCA-transformed set (17 principal components)

MODELING STRATEGY AND EXPERIMENTAL DESIGN

Model Implementation

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Model Evaluation & Comparison

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Hyperparameter Tuning
RandomizedSearchCV on top models, optimize F1

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Model Selection
Best F1 + PR-AUC

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Threshold Optimization
Threshold tuning (0.1 to 0.9), maximize F1

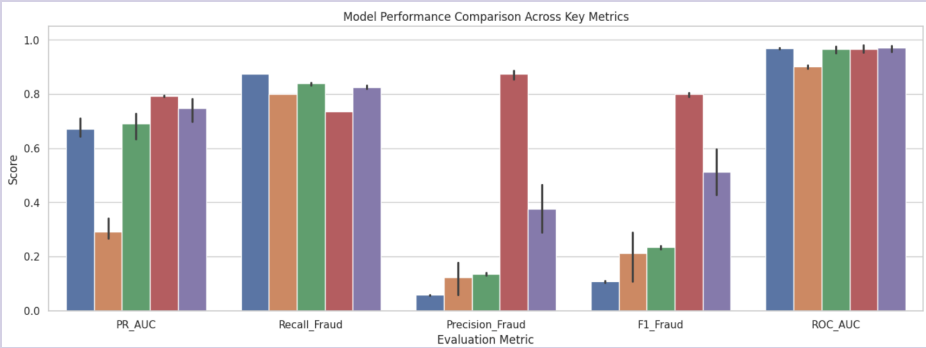
↓

Final Model

Model	Type	Rationale
Logistic Regression (LR)	Linear	Baseline model for comparison
Decision Tree (DT)	Tree-Based	Benchmark for non-linear models
Random Forest (RF)	Ensemble (Bagging)	Baseline ensemble model
Gradient Boosting (GB)	Ensemble (Boosting)	Test for boosting behavior
XGBoost (XGB)	Advanced Boosting	High potential for optimization

Metric	Strengths in Fraud Context	Rationale
Recall (Fraud)	Minimizes missed fraud	Critical for risk mitigation
Precision (Fraud)	Reduces false alarms	Evaluation of operational efficiency
F1-Score	Captures recall-precision trade-off	Primary optimization metric
PR-AUC	Robust to imbalance	Model ranking metric
ROC-AUC	Benchmark for overall separability	Secondary reference

MODEL IMPLEMENTATION AND EVALUATION



LR DT RF GB XGB

Findings per Model

- **LR:** High recall, lowest precision → high operational burden
- **DT:** High recall, weakest overall performance
- **RF:** Best untuned balance → strong potential for tuning
- **GB:** Good PR-AUC, low precision → high operational burden
- **XGB:** Strong recall, good PR-AUC → high tuning potential

	Dataset_Type	Model	Feature_Set	Recall_Fraud	Precision_Fraud	F1_Fraud	ROC_AUC	PR_AUC
0	2	Random Forest	PCA Features	0.7368	0.8537	0.7910	0.9531	0.7960
1	0	Random Forest	Selected Features	0.7368	0.8861	0.8046	0.9642	0.7940
2	1	Random Forest	Full Features	0.7368	0.8861	0.8046	0.9802	0.7907
3	1	XGBoost	Full Features	0.8316	0.4647	0.5962	0.9762	0.7827
4	0	XGBoost	Selected Features	0.8211	0.3768	0.5166	0.9786	0.7608
5	0	Gradient Boosting	Selected Features	0.8421	0.1368	0.2353	0.9752	0.7276
6	1	Gradient Boosting	Full Features	0.8316	0.1411	0.2412	0.9754	0.7150
7	1	Logistic Regression	Full Features	0.8737	0.0556	0.1046	0.9697	0.7109
8	2	XGBoost	PCA Features	0.8211	0.2889	0.4274	0.9568	0.6981
9	0	Logistic Regression	Selected Features	0.8737	0.0590	0.1105	0.9714	0.6609
10	2	Logistic Regression	PCA Features	0.8737	0.0601	0.1125	0.9671	0.6444
11	2	Gradient Boosting	PCA Features	0.8421	0.1311	0.2270	0.9501	0.6343
12	2	Decision Tree	PCA Features	0.8000	0.0582	0.1086	0.8954	0.3417
13	1	Decision Tree	Full Features	0.8000	0.1384	0.2360	0.9074	0.2677
14	0	Decision Tree	Selected Features	0.8000	0.1776	0.2906	0.9012	0.2668

Selected Models for Hyperparameter Tuning

Model	Feature Set	Rationale
Random Forest	PCA	Best untuned performance
Random Forest	Full	Second best performance
XGBoost	Full	Stronger recall vs. RF models

HYPERPARAMETER TUNING

Tuning Strategy

- **RandomizedSearchCV**: 10 parameter combinations using 3-fold stratified CV
- **StratifiedKFold** to preserve class distribution across folds
- **Optimized for F1-Score** to balance recall (minimize fraud) and precision (operational efficiency)

Summary of Performance Metrics – Before vs. After Tuning

	Model	Feature	Metric	Before	After	Δ
0	XGBoost Tuned	Full Features	F1	0.5962	0.8492	0.2530
1	XGBoost Tuned	Full Features	Recall	0.8316	0.8000	-0.0316
2	XGBoost Tuned	Full Features	Precision	0.4647	0.9048	0.4401
3	XGBoost Tuned	Full Features	PR-AUC	0.7827	0.8162	0.0335
4	Random Forest Tuned	Full Features	F1	0.8046	0.8000	-0.0046
5	Random Forest Tuned	Full Features	Recall	0.7368	0.7579	0.0211
6	Random Forest Tuned	Full Features	Precision	0.8861	0.8471	-0.0390
7	Random Forest Tuned	Full Features	PR-AUC	0.7907	0.7708	-0.0199
8	Random Forest Tuned	PCA Features	F1	0.7910	0.8023	0.0113
9	Random Forest Tuned	PCA Features	Recall	0.7368	0.7474	0.0106
10	Random Forest Tuned	PCA Features	Precision	0.8537	0.8659	0.0122
11	Random Forest Tuned	PCA Features	PR-AUC	0.7960	0.7945	-0.0015

Selected Model

Tuned XGBoost – Full Features

- **Highest PR-AUC:** Best performance across classification thresholds in an imbalanced setting
- **Highest F1-Score:** Strong balance between fraud detection and operational efficiency (low false positives)

Rank 1: Tuned XGBoost – Full Features Model

- Best overall performance
- Significant improvement in precision with minimal recall reduction

Rank 3: Tuned Random Forest – Full Features Model

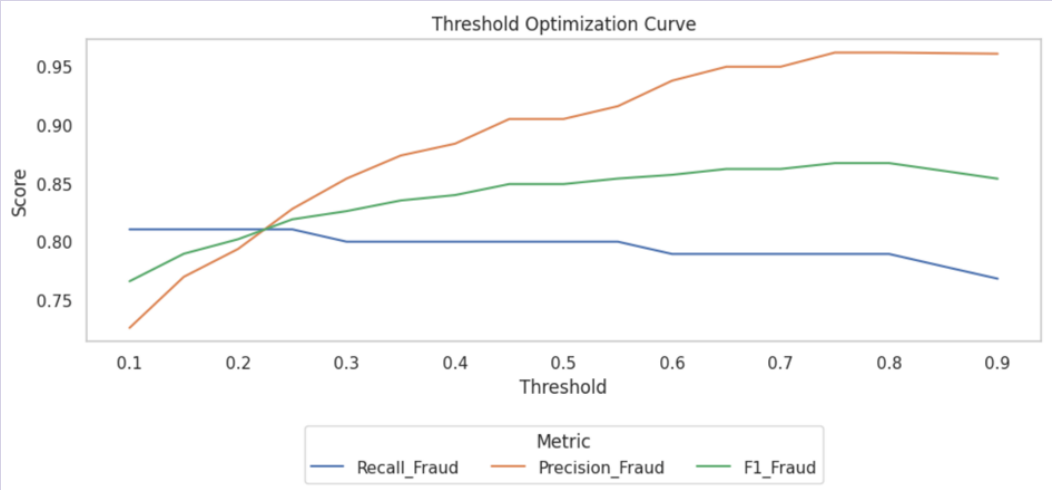
- Weakest performance among tuned models
- Slight increase in recall offset by slight precision reduction

Rank 2: Tuned Random Forest – PCA Features Model

- Almost no change after tuning → baseline already strong
- Negligible PR-AUC change, slight improvement in recall & precision

MODEL FINALIZATION

Threshold Optimization



Optimal Threshold = 0.75

- More conservative model (fewer false positives)
- Slightly more missed fraud cases (reduced recall)
- Better balance of recall-precision trade-off

Model	Threshold = 0.75	Threshold = 0.50	Rationale
Precision	0.9615	0.9048	Significant improvement
Recall	0.7895	0.8000	Slight trade-off
F1-Score	0.8671	0.8492	Better overall balance

Final Model

Tuned XGBoost, Full Feature Set, Threshold = 0.75

High precision

More reliable fraud alerts → Reduced operational burden and customer friction

Slightly lower recall

Some undetected fraud → Need complementary mechanisms to manage missed fraud

```
Confusion Matrix:
[[56648     3]
 [   20    75]]
```

MODEL LIMITATIONS

Model Performance Limitations

- False negatives → Missed fraud transactions
- False positives → Operational burden, customer friction

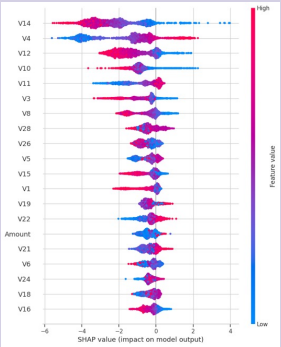
	Outcome_Type	Count	Percentage
0	TN	56648	99.8273
1	TP	75	0.1322
2	FN	20	0.0352
3	FP	3	0.0053

Interpretability and Transparency Limitations

- Key predictors are PCA-transformed → Explainable but not actionable
- SHAP explains model behavior, but not business meaning
- Limited ability to translate model outputs into actionable insights

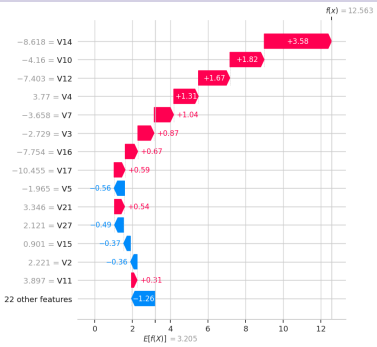
Global Explainability

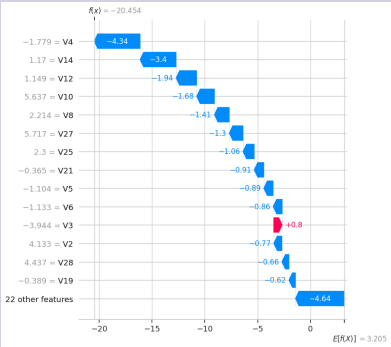
Key variables strongly influence predictions.



Local Explainability

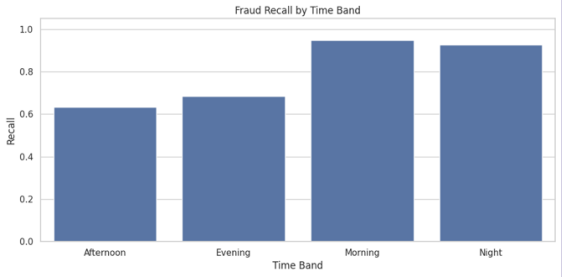
Individual decisions are explainable at the transaction level.

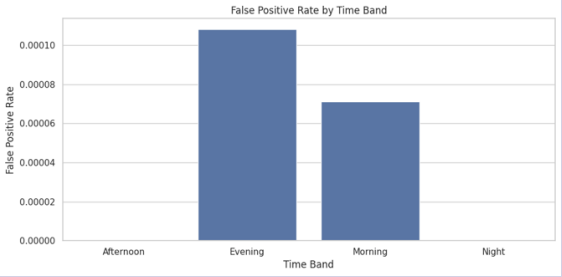




Fairness and Subgroup Limitations

- Uneven fraud detection across time bands
- Threshold adjustment did not resolve subgroup disparities
- Unobserved demographic groups may also show disparities





RISK MITIGATION AND ETHICAL AI

Priority	Risk Area	Mitigation	Ethical AI Considerations
High	Missed fraudulent transactions	Feature engineering	**Transparency:** Document model limitations
High	Uneven fraud detection performance across transaction segments	▪ Feature engineering ▪ Regular fairness monitoring	**Fairness Monitoring:** Track subgroup performance
High	Limitation of threshold-based optimization	Multi-stage or context-aware decision framework	**Balanced Decisioning:** Avoid systemic bias
High	Need for continuous governance and fairness monitoring	Metrics monitoring system across relevant segments	**Auditability:** Maintain logs and model tracking
Medium	Operational impact of false positives	Tiered decision framework	**Customer Impact:** Reduce friction
Medium	Limited business interpretability of anonymized and transformed features	▪ Interpretable monitoring features ▪ Business-facing explanation layers	**Human Oversight:** Enable explainability